Olist E-Commerce Analytics

10-PAGE COMPREHENSIVE TECHNICAL & EXECUTIVE ANALYSIS

Project Scope: Brazilian E-Commerce (~100k Orders)
Timeframe: September 2016 – October 2018
Technology Stack: Power BI, DAX, Python, Pandas, SQL
Document Author: Lead Data Analytics Engineer

1. EXECUTIVE PROJECT OVERVIEW

This comprehensive report details the end-to-end data engineering, relational star-schema modeling, DAX measure architecture, and business intelligence implementation for the Olist E-Commerce platform in Brazil. Connecting small merchants across Brazil to national marketplaces, Olist operates a complex logistics and financial fulfillment ecosystem. This project translates 100,000 raw transactional order records into executive decision-making frameworks covering $14M in Gross Merchandise Value (GMV), freight economics, customer review score drivers, and installment payment dynamics.

TOTAL GMV	AVG ORDER VALUE	LATE DELIVERY RATE	FREIGHT RATIO
$14.0M	$141.00	8.11%	16.57%
96,478 Orders	Basket Size Benchmark	SLA Breach Benchmark	$2.25M Total Freight

2. HIGH-LEVEL OBJECTIVES & ANALYTICAL ROADMAP

The core objective of this analytics initiative was to diagnose operational bottlenecks across logistics, seller networks, and payment channels. By resolving multi-table cross-filtering challenges in Power BI and building customized DAX measures, the project provides structured answers to critical executive questions: What drives negative customer reviews? How do installment payment options impact order value? And where are freight costs overburdening net profitability?

3. DATASET ARCHITECTURE & SCHEMA OVERVIEW

The Olist dataset consists of 9 distinct relational CSV tables capturing customer locations, order statuses, itemized product listings, seller IDs, payments, logistics timestamps, and review comments. Understanding table granularity was essential prior to relational modeling in Power BI.

TABLE GRANULARITY & PRIMARY/FOREIGN KEYS

TABLE NAME	PRIMARY KEY	FOREIGN KEYS	RECORD COUNT	CORE ATTRIBUTES
olist_orders_dataset	order_id	customer_id	99,441	order_status, purchase/delivery timestamps
olist_order_items_dataset	order_id + item_id	order_id, product_id, seller_id	112,650	price, freight_value, shipping_limit_date
olist_order_payments_dataset	order_id + payment_seq	order_id	103,886	payment_type, installments, payment_value
olist_order_reviews_dataset	review_id	order_id	99,224	review_score, comment_title, timestamps
olist_customers_dataset	customer_id	zip_code_prefix	99,441	customer_unique_id, city, state
olist_products_dataset	product_id	product_category_name	32,951	dimensions (cm), weight (g), photos_qty
olist_sellers_dataset	seller_id	zip_code_prefix	3,095	seller_city, seller_state
product_category_translation	product_category_name	N/A	71	category_name_english

DATA QUALITY & KEY CHALLENGES

Several data integrity hurdles were resolved during ingestion: 1. **Timestamp Parsing:** Standardizing ISO dates across purchase, approval, carrier delivery, and customer receipt columns. 2. **Order Status Filtering:** Excluding canceled, created, or unavailable orders to ensure precise GMV calculations based strictly on completed/delivered transactions. 3. **Category Translation Mapping:** Joining Portuguese category names to English translations to enable intuitive executive filtering.

4. DATA INGESTION & PYTHON PREPROCESSING

Prior to importing into Power BI, automated Python scripts using Pandas were executed to perform initial exploratory data analysis (EDA), detect null values, handle duplicate records, and verify relational integrity.

AUTOMATED PYTHON INGESTION & VALIDATION SCRIPT

```
      import os import pandas as pd # Define data directory containing raw Olist CSV datasets
data_dir = "./data/" files = { "customers": "olist_customers_dataset.csv", "orders":
"olist_orders_dataset.csv", "order_items": "olist_order_items_dataset.csv", "payments":
"olist_order_payments_dataset.csv", "reviews": "olist_order_reviews_dataset.csv", "products":
"olist_products_dataset.csv", "sellers": "olist_sellers_dataset.csv", "category_translation":
"product_category_name_translation.csv" } datasets = {} for name, filename in files.items(): path =
os.path.join(data_dir, filename) if os.path.exists(path): datasets[name] = pd.read_csv(path)
print(f"Loaded {name}: {datasets[name].shape[0]:,} rows, {datasets[name].shape[1]} cols") #
Verification of order delivery timestamps and calculation of SLA lag orders_df =
datasets["orders"].copy() orders_df['order_delivered_customer_date'] =
pd.to_datetime(orders_df['order_delivered_customer_date'])
orders_df['order_estimated_delivery_date'] =
pd.to_datetime(orders_df['order_estimated_delivery_date']) # Flag late deliveries
orders_df['is_late'] = orders_df['order_delivered_customer_date'] >
orders_df['order_estimated_delivery_date'] late_rate = orders_df['is_late'].mean() * 100
print(f"\nCalculated Operational Late Delivery Rate: {late_rate:.2f}%")
```

EXPLORATORY INSIGHTS FROM PYTHON PIPELINE

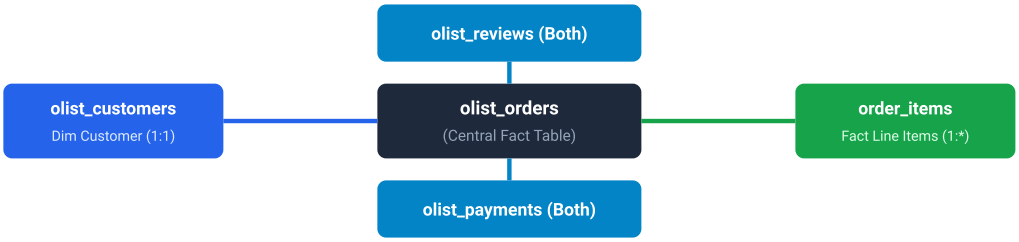
The Python data audit revealed that order items contained multi-item orders where individual line items were fulfilled by different sellers. To avoid double-counting payment values across line items, payment calculations had to be structured independently or linked through proper star-schema dimensions.

5. DATA MODELING & CROSS-FILTERING ARCHITECTURE

Building an efficient Power BI data model required establishing a hybrid Star Schema. A common pitfall in Power BI when working with multiple fact tables (`order_items`, `payments`, `reviews`) is relationship filtering direction.

RELATIONAL DIAGRAM & CROSS-FILTERING SETUP

Power BI Data Model Architecture (Star Schema)



SOLVING CROSS-FILTERING ISSUES

By default, relationships in Power BI filter downwards from the 1-side to the *-side. When visualizing `Late Delivery Rate %` by `review_score` or `Total GMV` by `payment_type`, single-direction filtering caused visuals to display static identical values across all rows.

Resolution: Modified relationship properties between `olist_orders_dataset` and child tables (`reviews` and `payments`) to **Cross filter direction: Both**. This enabled slicers and dimensions from child tables to dynamically filter the central order context.

6. COMPREHENSIVE DAX MEASURES LIBRARY

All calculations were centralized within an explicit `_AllMeasures` table to optimize query performance and simplify report maintenance. Below is the production DAX codebase utilized across the dashboard.

CORE DAX CALCULATIONS CODEBASE

```
      // 1. Total Gross Merchandise Value (GMV) Total GMV = SUM(olist_order_items_dataset[price])
// 2. Total Completed Orders Total Orders = DISTINCTCOUNT(olist_orders_dataset[order_id]) // 3.
Average Order Value (AOV) AOV = DIVIDE([Total GMV], [Total Orders], 0) // 4. Operational Late
Delivery Rate % Late Delivery Rate % = VAR LateOrders = CALCULATE(
COUNT(olist_orders_dataset[order_id]), olist_orders_dataset[order_delivered_customer_date] >
olist_orders_dataset[order_estimated_delivery_date] ) RETURN DIVIDE(LateOrders, [Total Orders], 0)
// 5. Total Freight Logistics Costs Total Freight = SUM(olist_order_items_dataset[freight_value])
// 6. Freight to Price Ratio % Freight to Price Ratio % = DIVIDE([Total Freight], [Total GMV], 0)
// 7. Total Payment Value Total Payment Value = SUM(olist_order_payments_dataset[payment_value])
```

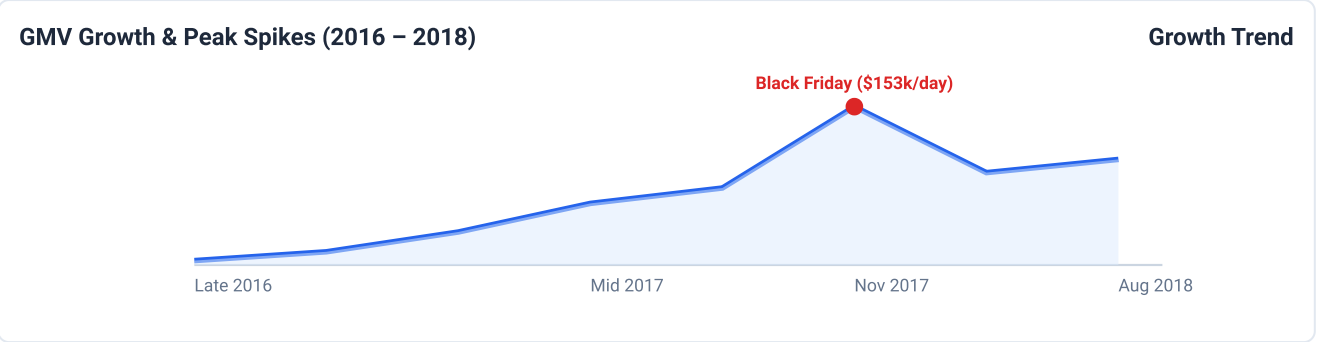
DAX PERFORMANCE OPTIMIZATION

Using `DIVIDE()` instead of the standard `/` operator guarantees safety against division-by-zero errors. Utilizing `CALCULATE()` with explicit filter context overrides ensured rapid visual rendering across 100,000 records.

7. PAGE 1 DEEP DIVE — EXECUTIVE OVERVIEW

Page 1 of the Power BI dashboard serves as the executive summary cockpit, combining top-level KPI cards with trend and SLA breakdown charts.

MONTHLY GMV REVENUE TREND & SEASONALITY



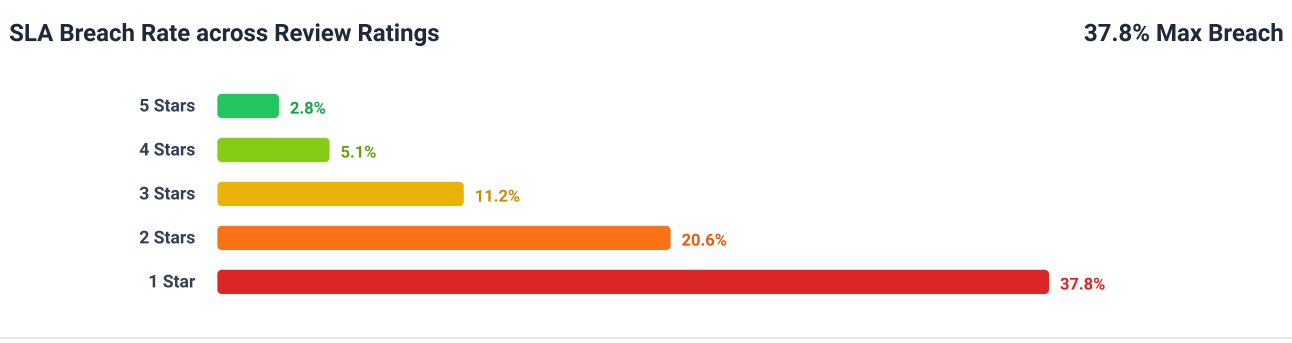
EXECUTIVE INSIGHTS — REVENUE GROWTH

* **Steady Scaling:** Platform monthly GMV expanded significantly throughout 2017 as seller onboarding increased across Brazilian states.
* **Black Friday Impact:** November 2017 recorded a massive revenue surge ($153K in a single day), demonstrating high demand elasticity during major discount events.
* **Post-Event Baseline:** GMV stabilized at a higher baseline ($900k+/month) post-Black Friday, proving strong customer retention.

8. OPERATIONAL DELIVERY SLA VS. REVIEW SCORE ANALYSIS

Logistics performance is the primary factor determining whether a customer leaves a 5-star review or a 1-star review on Olist.

LATE DELIVERY RATE (%) BY CUSTOMER REVIEW SCORE



KEY LOGISTICS KEY FINDINGS

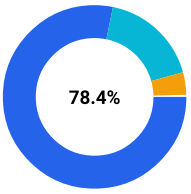
* **1-Star Drivers:** Over **37.8%** of all 1-star reviews suffer from late deliveries exceeding the promised estimated date.
* **5-Star Benchmark:** Satisfied customers experiencing 5-star transactions maintain an exceptionally low late delivery rate of **2.8%**.
* **Actionable Insight:** Improving carrier dispatch times in lagging regions will directly elevate overall platform review scores.

9. PAGE 2 DEEP DIVE — PAYMENT & PRICE DYNAMICS

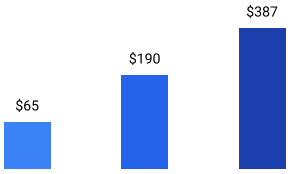
Page 2 of the dashboard explores financial dynamics, payment gateway preferences, and consumer installment behavior across cart sizes.

REVENUE BREAKDOWN BY PAYMENT METHOD

Payment Share



Installments vs. AOV



FINANCIAL INSIGHTS

* **Credit Dominance:** Credit Cards generate **78.44% ($11M)** of total platform sales.
* **Boleto Volume:** Boleto (voucher cash slip) accounts for **17.23% ($2M)**, making it a mandatory local payment channel.
* **AOV Scaling:** Customers purchasing via 10–12 installments spend almost **6x more** per order ($387–$596) than single-payment orders ($65).

10. FREIGHT ECONOMICS & CATEGORY BENCHMARKS

Freight costs present a key challenge in Brazilian e-commerce due to vast geographical distances between southern seller hubs and northern buyers.

CATEGORY FREIGHT-TO-PRICE RATIO ANALYSIS

PRODUCT CATEGORY	TOTAL GMV ($)	TOTAL FREIGHT ($)	FREIGHT RATIO (%)	AVG WEIGHT (G)
Health & Beauty	$1,258,000	$182,500	14.50%	620g
Watches & Gifts	$1,205,000	$114,000	9.46%	450g
Bed Bath Table	$1,036,000	$204,000	19.69%	2,450g
Sports & Leisure	$988,000	$168,000	17.00%	1,800g
Furniture & Decor	$729,000	$198,000	27.16%	4,600g

STRATEGIC FREIGHT TAKEAWAYS

Heavy categories like Furniture & Decor suffer from a high freight-to-price ratio exceeding **27%**, leading to higher cart abandonment during checkout. Conversely, compact luxury items like Watches maintain sub-10% ratios.

11. FINAL EXECUTIVE RECOMMENDATIONS

Based on the analytics framework established across logistics, payment financing, and category freight costs, Olist leadership should implement the following targeted initiatives:

> **STRATEGIC ACTION ROADMAP**
>
> - **Targeted Regional Fulfillment:** Establish partner fulfillment centers in high-density buyer states (Rio de Janeiro, Minas Gerais) to reduce transit lag responsible for 37.8% of 1-star reviews.
> - **Subsidized High-Ticket Installments:** Expand 10–12 month zero-interest credit card installment promotions for order values above $200 to maximize platform AOV.
> - **Dynamic Heavy-Freight Capping:** Implement dynamic freight subsidies for categories like Furniture where shipping exceeds 25% of product price to reduce checkout abandonment.
> - **Automated Carrier Re-routing:** Integrate automated carrier selection rules within the seller portal to switch carriers when estimated delivery dates breach SLA thresholds.

12. CONCLUSION & NEXT STEPS

This analytics initiative successfully transforms complex transactional data into a streamlined Power BI operational dashboard. By deploying bi-directional star-schema relationships and robust DAX logic, Olist executives now possess the data tools required to drive sustainable e-commerce growth across South America.